DIGITAL EGYPT PIONEERS INITIATIVE (DEPI) – DATA ANALYSIS TRACK

HR Data Analysis

Graduation Project Documentation • Team: Kareem (Team Leader), Youssef, Arsany, George, Abdallah

1. PROJECT PLANNING & MANAGEMENT

1.1 Project Idea & Scope

Project Idea: This project develops an interactive Human Resources (HR) Data Analysis Dashboard. As a team of junior data analysts, we take raw HR data stored in Excel, clean and transform it with Power Query, build a relational data model, and write DAX measures in Power BI. The final output is a visual dashboard that helps HR professionals track key employee metrics with ease.

Scope: The project covers importing HR data from Excel, cleaning and transforming the data with Power Query, building a simple relational data model, writing DAX measures (for example, Attrition Rate and Total Headcount), and designing Power BI dashboards.

1.2 Project Goals

• Clean and transform raw HR Excel data into a reliable, analysis-ready format.
• Build an intuitive Power BI dashboard, powered by DAX, that tracks workforce demographics and employee turnover.
• Give stakeholders clear, easy-to-read visuals that support decision-making.

1.3 Team Roles & Responsibilities

Tasks are distributed according to each member's strengths in Excel, Power Query, DAX, and Power BI:

Team Member	Role	Responsibilities
Kareem	Team Leader & Coordinator	Data collection, presentation, documentation, and task management
Arsany	Data Preparation & Modeling	Data cleaning and data modeling
Abdallah	Dashboard Design	HR Overview page and its DAX, and GitHub submission
George	Dashboard Design	Demographics page and its DAX
Youssef	Dashboard & DAX	Departmental Breakdown page and DAX measures and its DAX

1.4 Timeline & Milestones

Week	Milestone	Owner
Week 1	Requirements gathering, BRD completion, data collection, and initial Excel data preparation	**Kareem, Arsany**
Week 2	Data cleaning and transformation using Power Query	**Arsany**
Week 3	Data modeling and DAX calculations in Power BI	**Arsany, Youssef**
Week 4	Dashboard design (HR Overview, Demographics, Departmental Breakdown), final documentation, presentation, and GitHub submission	**Abdallah, George, Youssef, Kareem**

2. LITERATURE REVIEW

Traditionally, HR departments have relied on static Excel spreadsheets to track employee data. Excel works well for basic data entry and simple calculations, but it lacks the interactive, drill-down capabilities that modern HR reporting demands. Manual updates are also time-consuming and prone to error, which makes it difficult to keep reports consistent as the data grows.

> **OUR APPROACH & COMPETITIVE ADVANTAGE**
> This project addresses traditional static reporting limitations by taking standard Excel data and elevating it with Power BI. Power Query automates data cleaning, and DAX provides dynamic metrics such as Attrition Rate. The result is an interactive, easy-to-use solution that requires minimal technical knowledge from the end user.

3. REQUIREMENTS GATHERING

3.1 Business Requirements Summary

The purpose of this project is to give the HR department a clear, automated Power BI dashboard that replaces manual Excel reporting. The business needs to quickly identify how many employees are leaving (attrition) and to understand the demographics of the current workforce.

3.2 Stakeholders & Their Needs

HR Manager (Primary Stakeholder)

Needs a high-level overview of total headcount, attrition rate, and active employees to report to upper management.

Recruitment Team

Needs to see which departments have the highest turnover so future hiring can be planned effectively.

Junior HR Analysts

Need an easy-to-refresh Power BI file where they only update the underlying Excel sheet, without rewriting complex formulas.

3.3 & 3.4 Technical Requirements

3.3 Functional Requirements

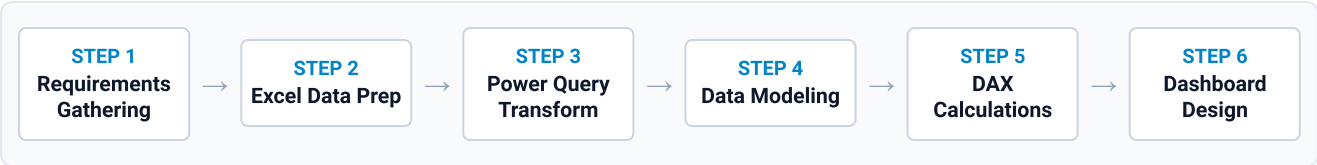
- **Overview Metrics (DAX):** Calculate Total Headcount, Active Employees, Terminated Employees, and Attrition Rate.
- **Demographic Analysis:** Visuals showing age distribution, gender ratio, and marital status.
- **Departmental Breakdown:** Bar charts showing employee count and attrition by department.
- **Interactivity:** Slicers (filters) for Year, Department, and Gender.

3.4 Non-Functional Requirements

- **Usability:** A clean, clear UI/UX suitable for non-technical HR staff.
- **Maintainability:** Power Query steps built so that data cleans itself automatically when a new Excel file is loaded.
- **Performance:** Simple, efficient DAX measures so the dashboard loads quickly.

4. PROJECT METHODOLOGY

The team follows a sequential workflow in which each stage feeds the next. Because the Power Query steps are built to be reusable, the dashboard refreshes automatically whenever the source Excel file is updated.



Core DAX Metrics Engine

TOTAL HEADCOUNT	ACTIVE EMPLOYEES	TERMINATED	ATTRITION RATE
Count of all recorded employee records	**Currently employed workforce total**	**Total employee departures**	**% turnover ratio calculated via DAX**

5. EXPECTED DELIVERABLES

- A cleaned and transformed HR dataset
- A Power BI data model with DAX measures (Total Headcount, Active Employees, Terminated Employees, Attrition Rate)
- Interactive dashboards covering workforce overview, demographics, and attrition
- Project documentation and a final presentation
- All project files submitted through GitHub

6. CONCLUSION

This project turns static HR spreadsheets into an interactive Power BI dashboard that lets HR stakeholders understand workforce composition and employee turnover at a glance. By combining Power Query for automated cleaning with DAX for dynamic metrics, the solution remains easy to maintain and accessible to non-technical users, giving the HR department a reliable foundation for reporting and decision-making.